

INDEPENDENT PEER-REVIEW PACKET

EC-008

Planar Collision-Gram Covariance Determinant

Cold Review Packet 4 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	4 of 20
Object ID	EC-008
Object class	Exact determinant identity
Exact statement	For the declared planar collision jets, covariance, Y-map, and c,s convention, $\det M = s^8 (c^2 + s^2) / (3c^2 + s^2)^{24}$.
Internal terminal label	CLOSED - EXACT PLANAR COLLISION-GRAM COVARIANCE IDENTITY
Terminal date	2026-07-21
Recommended expertise	Symbolic algebra / covariance matrices / Gaussian geometry

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Exact symbolic determinant routes and planted mutation sensitivity witness.
Prior review record	Independent determinant computations agreed after fixing the object card and coordinate convention.
Binding exclusions	No torus or finite-r transfer; no nondegeneracy theorem beyond this declared matrix.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No torus or finite-r transfer; no nondegeneracy theorem beyond this declared matrix.

4. Object-specific hostile checks

- Rebuild M from the declared source variables before computing its determinant.
- Factor the determinant independently.
- Apply the planted covariance or Y-map mutation and verify that the identity changes.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

GP-CLS-PROP-009-v1.0 — PROPOSED TERMINAL CLOSURE OF THE PLANAR COLLISION-GRAM COVARIANCE DETERMINANT

Artifact ID: GP-CLS-PROP-009-v1.0

Author: OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — additive terminal-closure proposal

Status: READY FOR HUMAN APPROVAL

Authority: none until Dylan M. Roy accepts or modifies the terminal statement

Canonical impact: NONE before approval

Queue item: EC-008

1. EXACT OBJECT TO BE CLOSED

Let the centered planar coalesced free jets

$$(a_{yy}, a_{xxy}, a_{xyy}, a_{yyy})$$

have covariance

$$\text{diag}(2, 2, 2, 6).$$

For declared collision coordinates c and s , define

$$\begin{aligned} Y1 &= -a_{xxy} c^2 s / 4 - a_{xyy} c s^2 / 4 - a_{yyy} s^3 / 12, \\ Y2 &= a_{xxy} c s + a_{xyy} s^2 / 2, \\ Y3 &= a_{yy} s. \end{aligned}$$

Let

$$M = \text{Cov}(Y1, Y2, Y3).$$

Then

M =

$$\begin{bmatrix} s^2(3c^4 + 3c^2s^2 + s^4)/24, & -cs^2(2c^2 + s^2)/4, & 0, \\ -cs^2(2c^2 + s^2)/4, & s^2(4c^2 + s^2)/2, & 0, \\ 0, & 0, & 2s^2 \end{bmatrix},$$

and the exact determinant is

$$\det M = s^8(c^2 + s^2)(3c^2 + s^2)/24.$$

Proposed terminal label:

CLOSED — EXACT PLANAR COLLISION-GRAM COVARIANCE IDENTITY

2. FIRST-PRINCIPLES DERIVATION

Because the four source jets are independent under the declared planar coalesced covariance, covariance is obtained by summing products of coefficients weighted by the source variances 2,2,2,6.

The exact diagonal terms are

$$\begin{aligned} M_{11} &= s^2 (3c^4 + 3c^2 s^2 + s^4) / 24, \\ M_{22} &= s^2 (4c^2 + s^2) / 2, \\ M_{33} &= 2s^2. \end{aligned}$$

The only nonzero off-diagonal entry is

$$M_{12} = M_{21} = -cs^2 (2c^2 + s^2) / 4.$$

Y3 uses only a_{yy} , while Y1 and Y2 use only a_{xxy} , a_{xyy} , and a_{yyy} , so

$$M_{13}=M_{23}=0.$$

The determinant therefore reduces to

$$\begin{aligned} \det M &= 2s^2 (M_{11} M_{22} - M_{12}^2). \end{aligned}$$

Exact expansion and factorization give

$$\begin{aligned} M_{11} M_{22} - M_{12}^2 &= s^6 (c^2 + s^2) (3c^2 + s^2) / 48, \end{aligned}$$

hence

$$\det M = s^8 (c^2 + s^2) (3c^2 + s^2) / 24.$$

No asymptotic argument, simulation, or unproved theorem is used.

3. EVIDENCE AND INDEPENDENT REVIEW

Primary executable source:

- C047R_verification_battery_01.py, Drive ID 1oUx_WO82T_3xp3jQX9FlbfTW0Sb9apk-, source SHA-256 149920344bd4f0ad7f1b0f89793a229894bf2bb31c853315322a0a097e676c76. The source derives Y1, Y2, Y3 from the declared cubic jet polynomial, constructs M from the exact source variances, verifies the full matrix, verifies the determinant factorization, and includes a planted variance perturbation.

Independent OpenAI re-execution:

- GP-AUD-011-v1.0, Drive ID 1i0jWKL1-hhe2yIQhlzQeK4MR51vXbLmpfrL2dYqUYgk, downloaded and hashed the exact source, ran it unmodified, and independently reconstructed the collision map and covariance determinant. It records the matrix and determinant as CONFIRMED-BY-REEXECUTION.

Negative control:

- Changing $\text{Var}(a_{yyy})$ from 6 to 5 changes M11 by exactly $-s^6/144$. GP-AUD-011 independently confirms this planted perturbation, showing that the symbolic test is sensitive to a load-bearing covariance change rather than merely returning a pass string.

Independent C047R acceptance:

- CW-RSP-012-v1.0, Drive ID 1q10Wg4YNj_5cF8a7kOPJhAtJJXzM9_Qo, accepts GP-AUD-011 in full and explicitly confirms the collision determinant and matrix entries.

Fresh OpenAI exact reconstruction:

- This closure proposal independently recomputes every covariance entry from the source coefficients and factors the 3×3 determinant through its 2×2 block.

4. LOAD-BEARING SCOPE LIMIT

This closure applies only to the declared PLANAR_COALESCED collision-map object with:

- source-jet order ($a_{yy}, a_{xxy}, a_{xyy}, a_{yyy}$);
- source covariance $\text{diag}(2, 2, 2, 6)$;
- the exact Y_1, Y_2, Y_3 definitions displayed above;
- the declared c, s coordinate convention.

It does NOT assert that the same matrix or determinant holds without modification for:

- an exact normalized-periodized finite- r torus law;
- a different free-jet ordering;
- a different collision coordinate map;
- correlated or differently normalized source jets;
- a different determinant or Palm-weight convention.

5. DEPENDENCIES AFFECTED

After approval, this closure permits the displayed matrix and determinant to be cited without reopening symbolic algebra whenever the object card matches exactly.

It supports:

- collision-map nondegeneracy checks for $s \neq 0$;
- verification-battery regression tests;
- planar local Kac–Rice and Gaussian-density calculations that use this exact covariance object;
- object-card validation preventing convention drift;
- future audits that require the exact determinant power s^8 and its positive factors.

6. DEPENDENCIES UNAFFECTED

This closure does not:

- • prove a complete Palm-law or Kac–Rice theorem;
- • establish endpoint dynamics, adjacency, selection, or capture;
- • close adjacency positivity, adjacency-to-one, or P0.2;
- • prove exact-torus or finite-r transfer;
- • verify probability tails or uniform integrability;
- • change Theorem B or the live q_0 theorem state;
- • promote a machine root or successor machine state;
- • validate the battery’s unrelated weak geometry assertion;
- • alter any historical source file.

7. TERMINAL RECORD AND REGISTER ACTIONS AFTER APPROVAL

After Dylan M. Roy approves or modifies the terminal statement:

1. Move EC-008 from Easy Closure Queue to Closure Log.
2. Record terminal label:

CLOSED — EXACT PLANAR COLLISION-GRAM COVARIANCE IDENTITY

3. Register object tags:

PLANAR_COALESCED

COLLISION_MAP_Y1_Y2_Y3

SOURCE_COVARIANCE_DIAG_2_2_2_6

DET_M_S8_FACTOR

NOT_EXACT_TORUS_FINITE_R

4. Cross-link C047R_verification_battery_01.py, GP-AUD-011, and CW-RSP-012.
5. Remove this identity from active re-derivation and routine review queues.
6. Preserve every source, audit, response, and correction artifact unchanged.

8. HUMAN APPROVAL REQUIREMENT

This item is mathematically closure-ready but remains a human terminal decision under OP-PROT-004.

Proposed approval statement:

I approve GP-CLS-PROP-009-v1.0. Close the exact planar collision-map covariance identity for the declared source jets, source covariance $\text{diag}(2,2,2,6)$, and Y_1, Y_2, Y_3 definitions, including $\det M = s^8(c^2 + s^2)(3c^2 + s^2)/24$. Preserve the distinction from exact-torus, finite-r, and differently normalized collision objects.

9. CORRECTION PATH

If a later audit finds an algebraic, coordinate, normalization, sign, or object-definition error, do not edit this document silently. File an additive erratum identifying the exact failed line, affected dependencies, and corrected terminal status.

END OF PROPOSED CLOSURE PACKAGE

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT PLANAR COLLISION-GRAM COVARIANCE IDENTITY
Closure date	2026-07-21
Evidence basis	Exact symbolic determinant routes and planted mutation sensitivity witness.
Independent review already recorded	Independent determinant computations agreed after fixing the object card and coordinate convention.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No torus or finite-r transfer; no nondegeneracy theorem beyond this declared matrix.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: https://docs.google.com/document/d/1LCkRwiIOc0ThpM1EpC9_1oHSZrLleYfT270KDWKVhIY/edit

Closure / terminal record: https://docs.google.com/document/d/1Zm65j9BZe_z3P_8qP7rfZukRU-IWUE8h8bDqXdDsIQ/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-009-v1.0 — Proposed terminal closure of the planar collision-Gram covariance determinant

Local source snapshot SHA-256 (DOCX export, transport checksum only):
c76dda418b4cdf353e55a4012f1b812c49493441ba01fe49223cadfc821627a4

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.